

$$e = \frac{P\phi}{60/N} = \frac{\phi PN}{60} \text{ voltz}$$

emf induced per parallel path $\frac{\phi PN}{60} \times \frac{Z}{A}$

$$e = \frac{\phi PN}{60} \times \frac{Z}{A}$$

$$e = \frac{\phi ZNP}{60A}$$

$$E_g = \frac{\phi ZNP}{60A} \text{ volts}$$

Gen/induced emf

$$E_g \propto \phi N$$

$$\frac{E_{g1}}{E_{g2}} = \frac{\phi_1 N_1}{\phi_2 N_2}$$

(OR)

$$\frac{E_{g2}}{E_{g1}} = \frac{\phi_2}{\phi_1} \times \frac{N_2}{N_1}$$

If N is constant

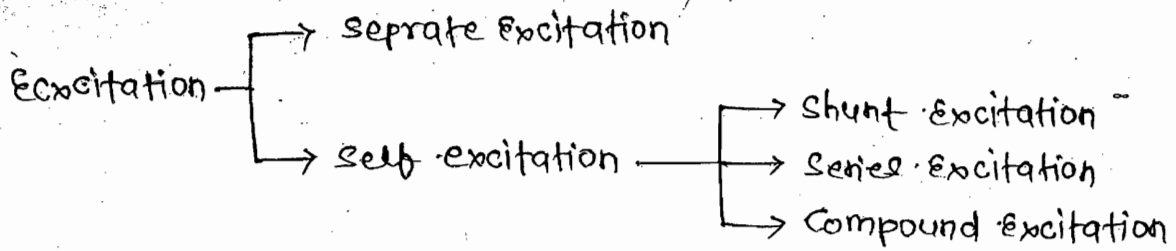
$$\frac{E_{g2}}{E_{g1}} = \frac{\phi_2}{\phi_1}$$

If ϕ is constant

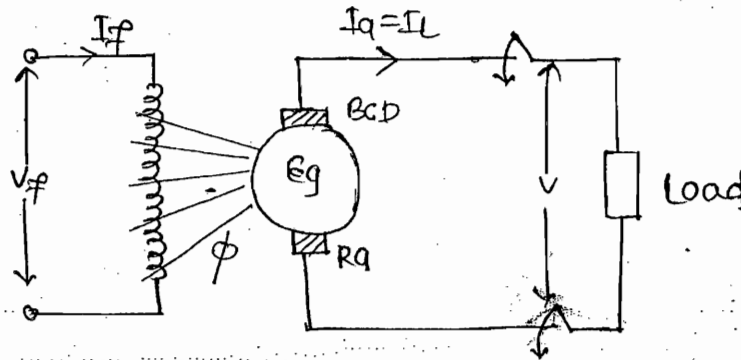
$$\frac{E_{g2}}{E_{g1}} = \frac{N_2}{N_1}$$

Classification of dc gen^r → * In order to control the m/c electro-magnets are preferred as poles which requires a dc vol. excitation across it

* The classification of gen^r is according to method of excitation:-



* Seprate Excitation →



* It requires a seprate dc source for its excitation.

* Its field & arm. wdg are isolated electrically.

* The terminal voltage (V) across the load doesn't affect its excitation.

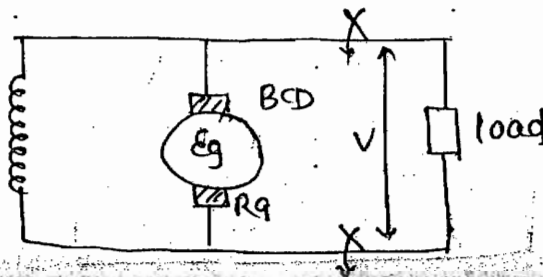
* Due to additional voltage source requirement it is rarely used gen^r around 5% dc gen^r are self excited only.

gen/Ind emf

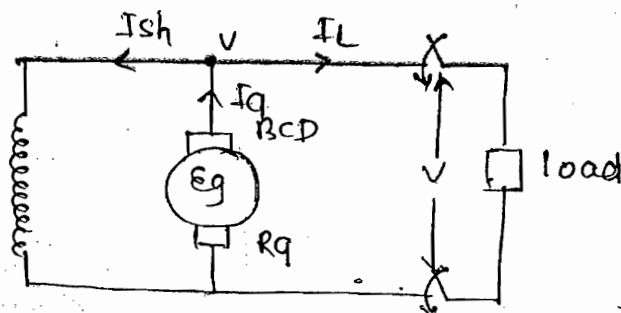
$$E_g - I_a R_a - BCD = V^{\text{Terminal } V}$$

$$I_a = I_L$$

* Self excitation → * The field wdg will be excited by its own arm. which requires some essential condⁿ starts with residual flux.



Shunt excitation →



$$E_g = I_f R_f - BCD = V$$

$$I_f = I_L + I_{sh}$$

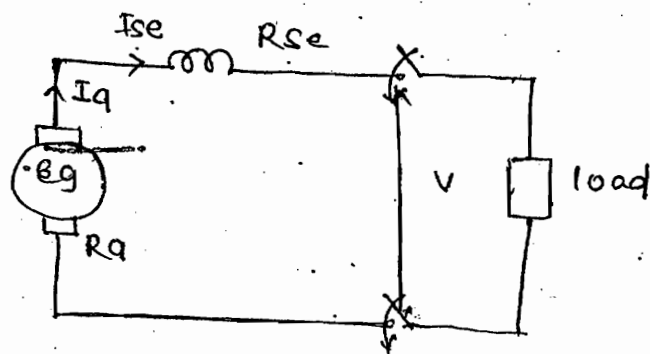
$$I_{sh} = \frac{V}{R_{sh}}$$

- * The field wdg is connected in parallel with arm. through brushes.
 - * The terminal vol. itself act as excitation.
 - * Shunt field turns are very large in no. due to its resistance is high with thin cond^r.
- Range 50-250 Ω

Loading any m/c = Reducing the ~~at~~ resistance

- * The field ~~turn~~ current remains approx. same from NL to rated value.
- * Consequently flux is approx. same in the operating region.
- * In this mode it is known as shunt gen^r & vol. operated field.

Series excitation →

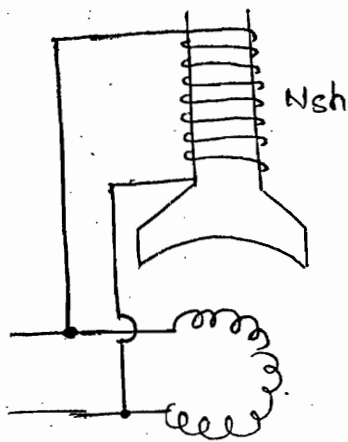


- * In this mode it is known as series gen^r & current operated field.

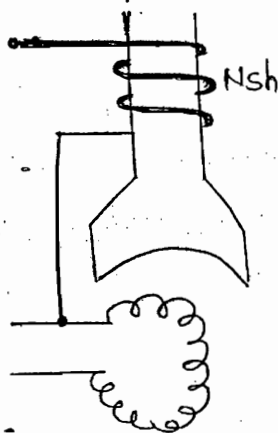
$$E_g = I_f (R_f + R_{se}) - BCD = V$$

$$I_f = I_L = I_{sh}$$

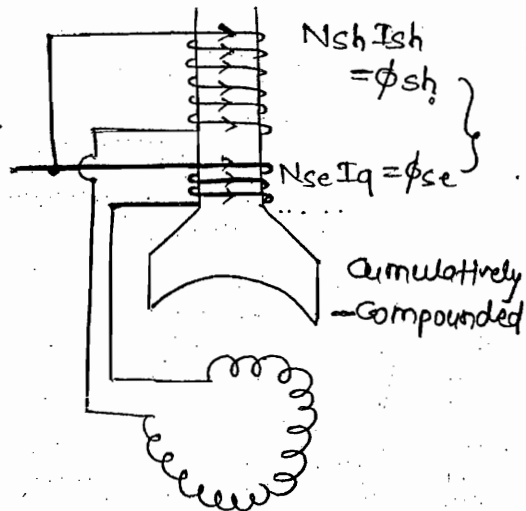
* The series field wdg contain less no. of turns with thick cond?



Shunt



Series



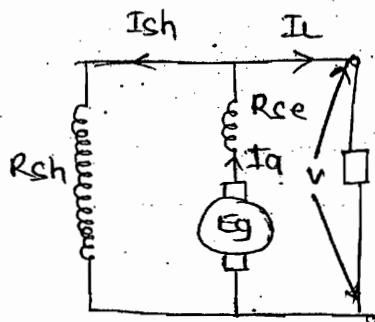
Compound

* Shunt flux always dominate series flux

* If series field wdg is reversed than $\phi_{sh} - \phi_{se}$: Differently

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Long shunt →

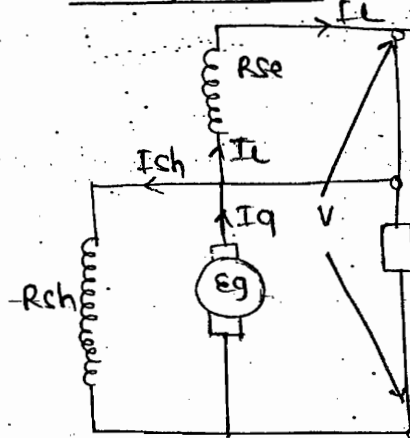


$$E_g = I_a(R_a + R_{se}) - BCD = V$$

$$I_a = I_L + I_{sh}$$

$$I_{sh} = V / R_{sh}$$

Short shunt →

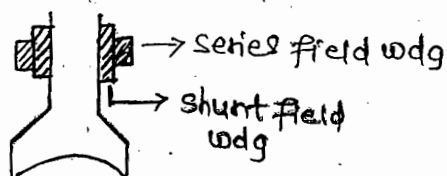


$$E_g - I_a R_a - I_L R_{se} - BCD = V$$

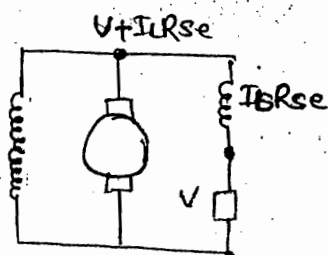
$$I_a = I_{sh} + I_L$$

$$I_{sh} = \frac{V + I_L R_{se}}{R_{sh}}$$

* There is no



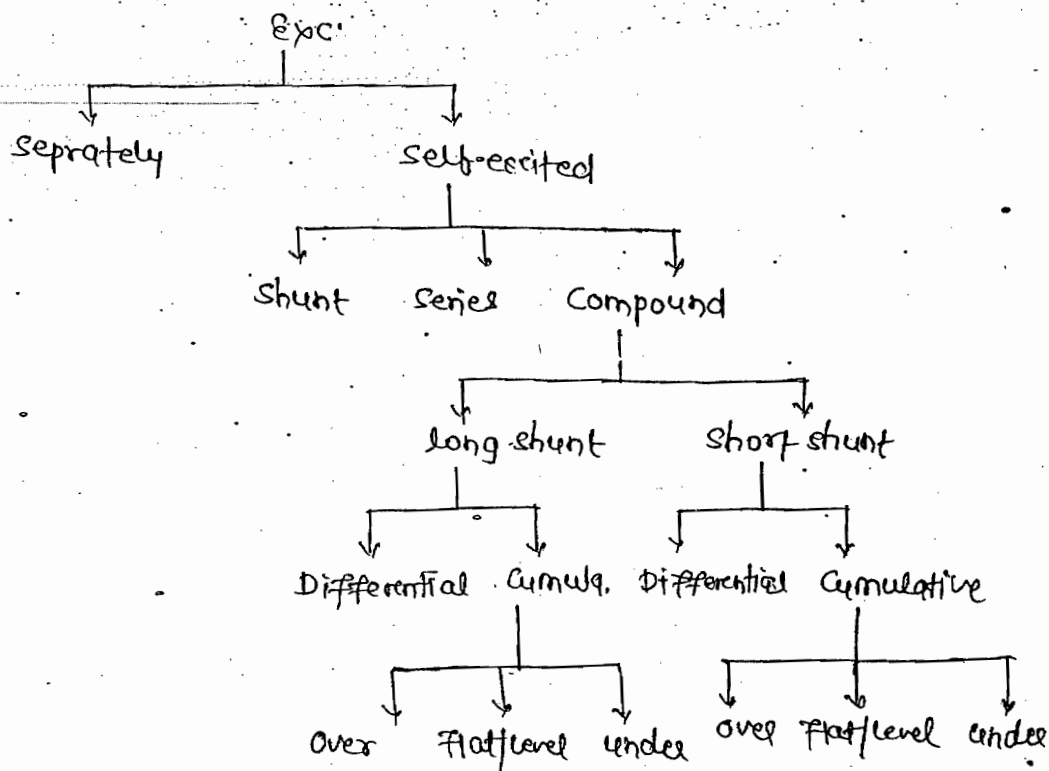
Short shunt



- * There is no distinctive diff. b/w long shunt & short shunt gen^r.
- * The induced emf in long shunt is slightly greater than that of short shunt ($I_a > I_w$)
- * If the series field wdg. is reversed than the cumulative mode becomes differential.

Cumulative currents in both field wdg same dirⁿ on the pole core. Therefore both fluxes add each other. and the net flux increases with load.

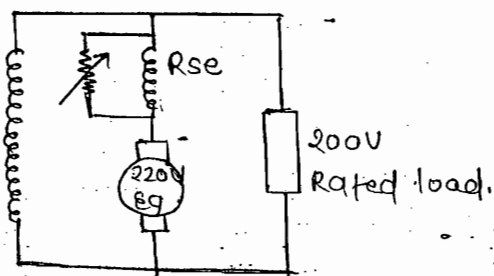
In differential the series flux oppose the shunt flux & the net flux decreases with load.



Compounding → * Adjusting the terminal voltage of cumulative compound gen^r by varying its series field ampere turns by connecting a diverter across its series field wdg.

* Depending on the degree of compounding a cumulative compound gen^r can act as under, flat, over compound gen^r.

Compounding is done at rated load. to adjust the terminal vol. below rated, exactly rated & above rated respectively.



* By varying the diverter resistance from 0 to max^m value a cumulative compound gen^r can be acted in 4 other modes.

- (i) $R_d = 0$ (shunt)
- (ii) $R_d \uparrow$ Basic cumulative
- (iii) $R_d \uparrow \uparrow$ Under
- (iv) $R_d \uparrow \uparrow \uparrow$ Flat/level
- (v) $R_d \uparrow \uparrow \uparrow \uparrow$ Over

* Compounding can't be done with differential compound gen^r as its series flux opposes shunt flux.

* The induced emf of flat compound gen^r is exactly equal to rated voltage at rated load.

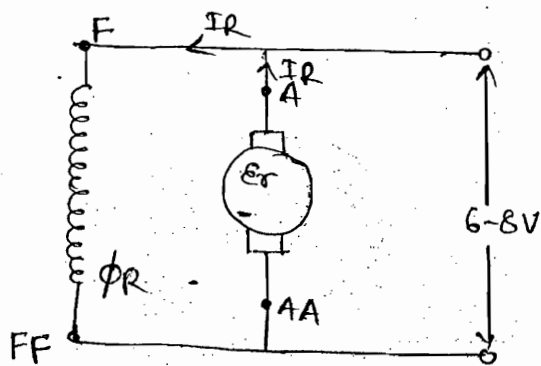
Since series flux is adjusted to compensate all the drops at rated load.

Voltage build up → [In self excited gen^r]

* In a separately excited gen^r there is a separate source to excite the field & produce flux.

* In a self excited gen^r there is no such source to excite the field wdg. Therefore in order to build up voltage in self excited gen^r the poles should contain the residual flux.

Shunt gen^r →

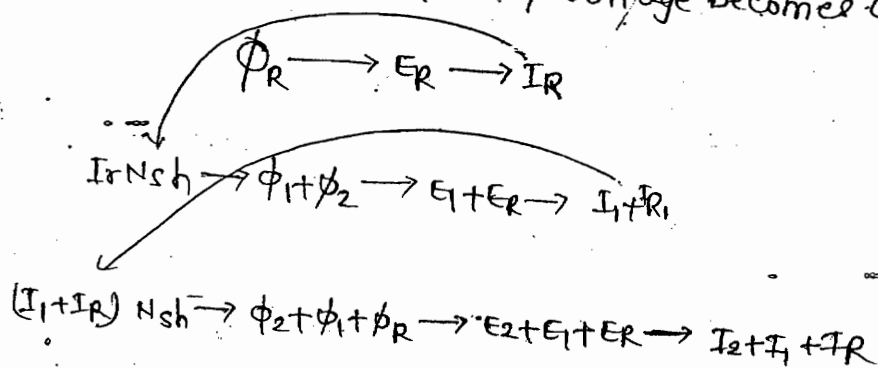


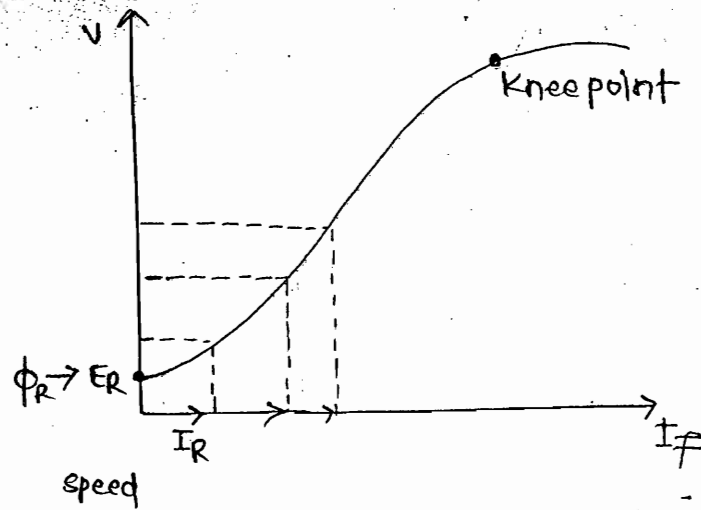
1) * Consider a shunt gen^r made to run at rated speed with residual flux in its poles. Small voltage is induced in the arm. (6-8V) which produces small current.

* If the gen^r is on NL, all the initial current will flow into field wdg. to produce initial mmf with field wdg turns.

2) * The field should be properly connected to the arm. in order to make initial mmf at the residual flux.

* It is a cumulative process as the current increases the flux out of the pole increased & the induced voltage also increases after saturation of poles even though the current increase. Flux doesn't increase. Consequently voltage becomes constant.





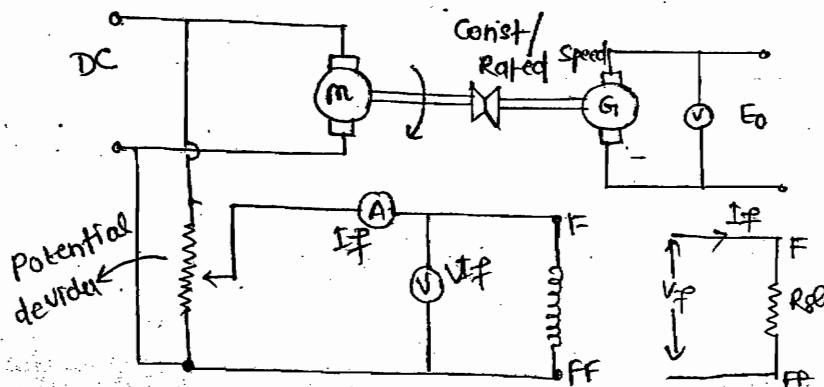
- (3.) The operating ~~cond~~ speed of the gen^r should be greater than critical speed.
- (4.) The resistance of field wdg should be less than critical resistance value.
- (5.) The best condⁿ for a shunt gen^r to build up voltage successfully is it should be on NL.
- (6.) In spite of all the above 5 condⁿ satisfied if the vol. doesn't build up it may be due to improper contact across brushes & commutator.

Determination of critical resistance & speed $\rightarrow$

OCC (open circuit c/s)
NLS (No load saturation)
Magnetisation c/s

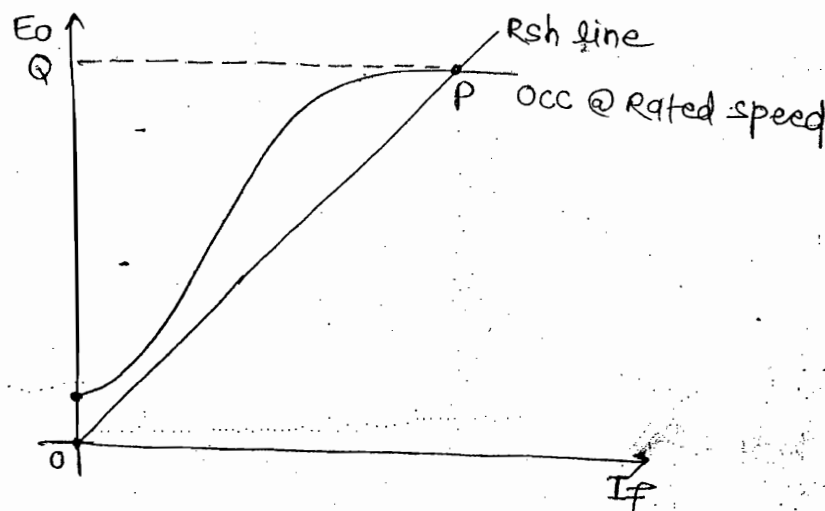
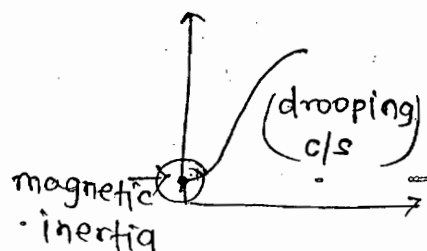
} E_o Vs I_f

* In order to plot OCC to determine R_c & N_c of a self exc shunt gen^r it requires to be separately excited.



S.N.	V_f	I_f	E_o
1	0	0	6.8
2	10	0.1	30
3	20	0.2	60
4	30	0.3	80

linear

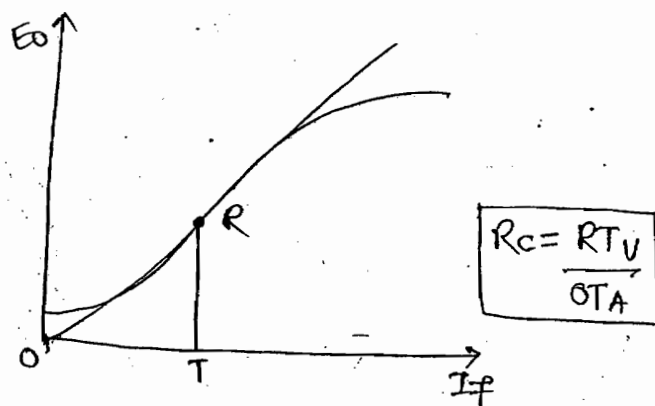


Critical Resistance → The resistance of the field wdg. about which the genr doesn't build up voltage ~~this is~~

Critical speed → The speed of genr below which the genr doesn't build up voltage.

Determination of critical resistance →

- * The field resistance line intercepts OCC at a point P.
- * Length of OQ in volts is the maxm emf induced in the genr.



Steps: (1) Draw the OCC line.

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(2) Plot a tangent through OCC.

Critical speed Determination →

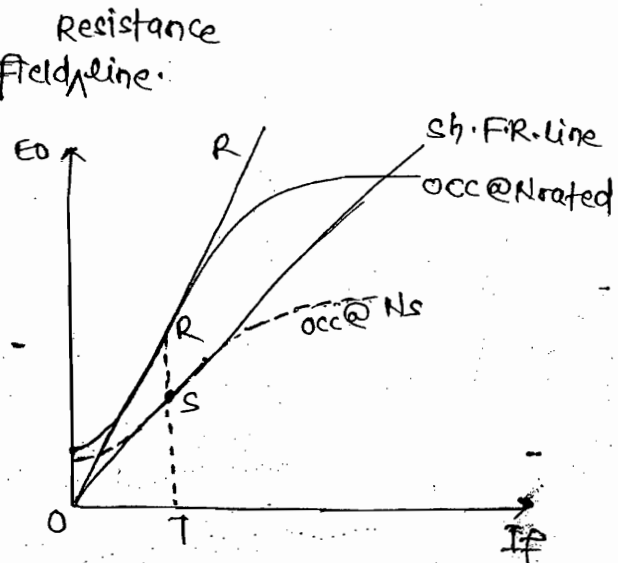
Steps → (1) Draw original shunt field line.

length of (ST) $\propto N_c$

length of (RT) $\propto N_{rated}$

$$\frac{ST}{RT} = \frac{N_c}{N}$$

$$N_c = \frac{ST}{RT} N$$



* The critical resistance of a given gen^r depends on the operating speed. From critical speed & above.

* It varies proportionally with speed.

* When the m/c is running at critical speed its field resistance value itself is critical resistance.

Eg. → A shunt gen^r building up vol. normally. If the field wdg is reversed & operated that

(a) Build up vol. normally

(b) Build up vol. with -ve polarity

(c) No build up of vol.

(d) GV across the arm.

Ans. → (c)

Eg. → Same above que. Direcⁿ of rotation of gen^r is reversed.
options same.

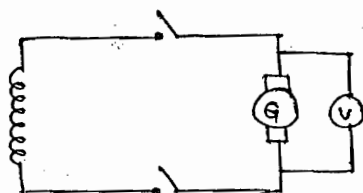
Ans. → (c) $-E_g \propto \phi(-N)$

eg. → If both direcⁿ of rotation as well as field wdg. is reversed.

$$E_g \propto (\phi)(-N) \text{ is } E_g (+ve)$$

Ans. (a)

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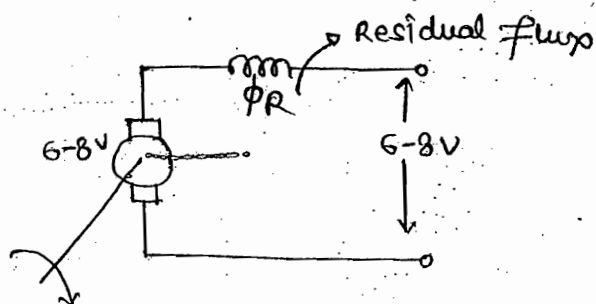


switch

Open Close

A.	8V	8V	1. Critical Resistance
B.	8V	12V	2. load resistance
C.	8V	0V	3. Field polarity

* Voltage build up in series gen^r →



- (1.) It requires residual flux.
- (2.) The terminals should be closed with some load. As the load current flows into the field wdg the initial mmf is considerably good value & no need of large series field wdg turns.
- (3.) The total resistance ($R_a + R_{se} + R_L$) should be less than its critical resistance.

$$(R_a + R_{se} + R_L) < R_c$$
- (4.) Its speed should be greater than critical speed.

Cumulative Compound gen^r →

* A compound gen^r is eq. to a shunt gen^r on NL. Therefore its vol. build up is eq. to shunt gen^r.